

HOTEL BOOKING DEMAND ANALYSIS

Exploratory Data Analysis (EDA) Report

Project: Hotel Booking Demand Analysis

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Flask

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Hotel Booking Demand Analysis using Python, Statistics, EDA and Flask Dashboard

Project Overview

The Hotel Booking Demand Analysis project focuses on analyzing hotel reservation data to understand customer booking behaviour, seasonal demand, cancellation patterns, pricing trends and hotel performance. Exploratory Data Analysis (EDA) was performed using Python libraries such as Pandas, NumPy, Matplotlib and Seaborn to identify important patterns, relationships and business insights. The findings from this analysis help hotel management improve occupancy, optimize pricing strategies, reduce cancellations and support data-driven decision making.

1. Monthly Booking Analysis



Observation

The monthly booking trend indicates clear variations in hotel reservations throughout the year. August records the highest number of bookings, while some months experience comparatively lower demand.

Interpretation

Booking demand changes according to seasonal travel, public holidays and vacation periods. Peak months generate significantly higher occupancy rates.

Business Insight

Hotels experience maximum customer demand during peak travel months. Understanding these seasonal trends enables hotel management to forecast occupancy, optimize staffing levels and maximize revenue.

Recommendation

- Increase room availability during peak seasons.
- Apply dynamic pricing strategies.
- Launch promotional campaigns during low-demand months.
- Improve workforce planning based on booking forecasts.

2. Hotel Distribution



Observation

The City Hotel receives a higher number of bookings than the Resort Hotel. Resort Hotels account for a comparatively smaller share of total reservations.

Interpretation

Customers prefer City Hotels because of easier accessibility, business travel, and urban tourism. Resort Hotels are mainly chosen for vacations and leisure trips.

Business Insight

City Hotels contribute a larger portion of total bookings and revenue. Resort Hotels experience more seasonal demand and lower occupancy during off-peak periods.

Recommendation

- Increase promotional campaigns for Resort Hotels during off-season periods.
- Continue dynamic pricing strategies for City Hotels during high-demand periods.
- Introduce vacation packages and family offers to improve Resort Hotel occupancy.

3. Season Distribution



Observation

Hotel bookings vary significantly across different seasons. Certain seasons experience considerably higher booking volumes, while others have comparatively lower customer demand.

Interpretation

Seasonal demand strongly influences hotel occupancy. Holiday periods, weather conditions and vacation schedules play an important role in customer booking behaviour.

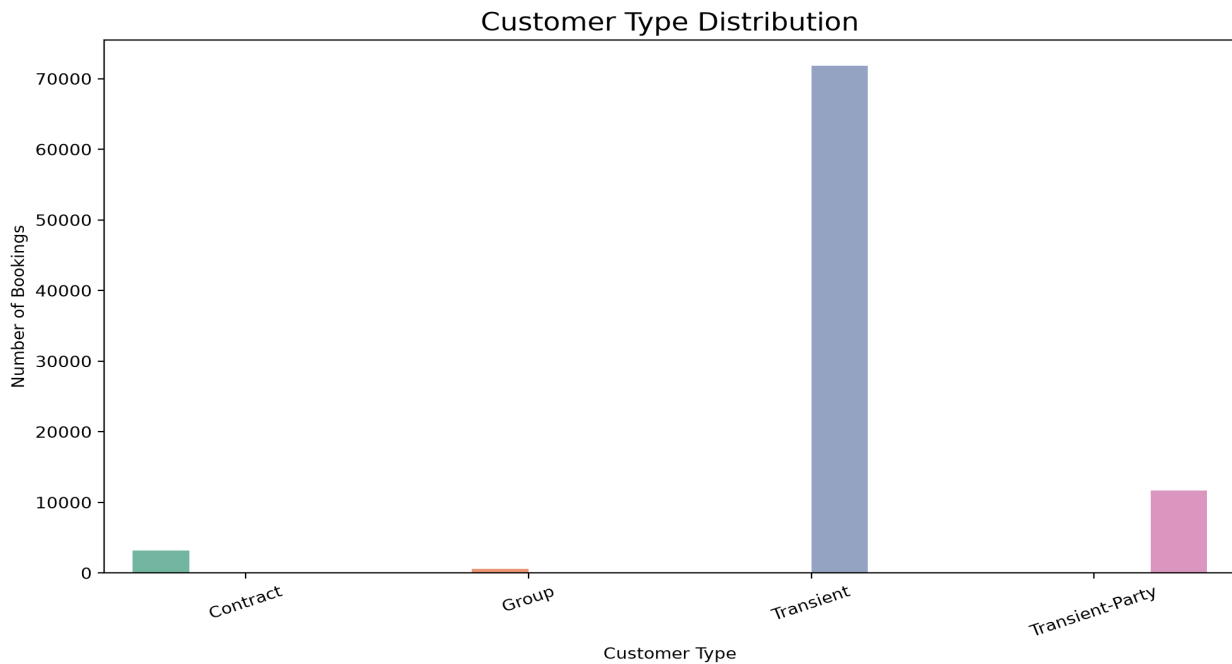
Business Insight

Understanding seasonal demand enables hotel management to optimize pricing strategies, allocate resources efficiently and prepare for fluctuations in customer arrivals.

Recommendation

- Increase room prices during high-demand seasons.
- Launch promotional offers during off-season periods.
- Plan staffing, inventory and operational activities according to seasonal booking trends.

4. Customer Type Analysis



Observation

The distribution of bookings varies among different customer types. Transient customers contribute the largest share of hotel bookings, while Contract, Group and Transient-Party customers represent comparatively smaller segments.

Interpretation

Most hotel reservations are made by individual travellers rather than organized groups or corporate contracts. This indicates that the business primarily depends on independent customers.

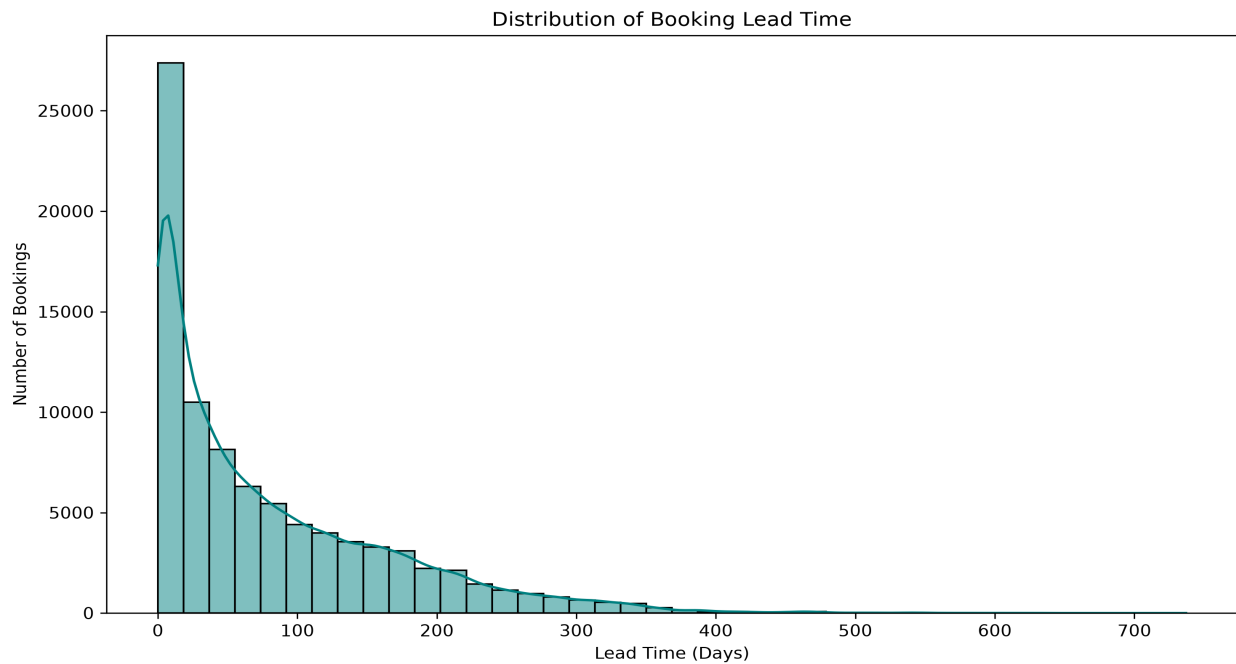
Business Insight

Customer segmentation helps hotels understand booking behaviour, develop targeted marketing campaigns and improve customer retention strategies.

Recommendation

- Introduce loyalty programs for frequent transient customers.
- Offer corporate discounts to increase contract bookings.
- Develop customized packages for group reservations.
- Personalize promotional campaigns based on customer type.

5. Lead Time Distribution



Observation

The lead time distribution is positively skewed. Most hotel bookings are made within a relatively short period before the arrival date, while only a small proportion of customers book several months in advance.

Interpretation

Customers generally prefer making reservations closer to their travel dates. Long lead-time bookings are less common but provide better visibility for occupancy planning.

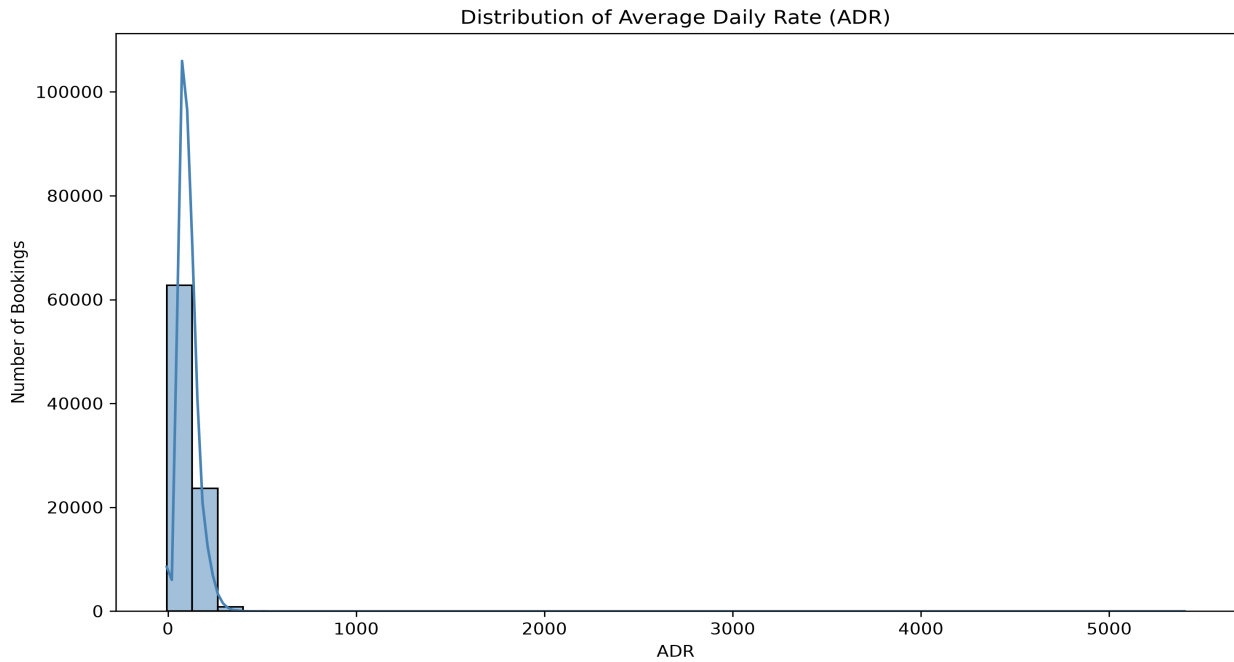
Business Insight

Lead time has a significant impact on hotel demand forecasting, pricing strategies and cancellation management. Monitoring booking patterns helps hotels optimize room allocation and revenue planning.

Recommendation

- Encourage early reservations through advance booking discounts.
- Apply dynamic pricing for last-minute bookings.
- Improve occupancy forecasting using historical lead-time trends.
- Monitor long lead-time reservations to reduce cancellation risk.

6. Average Daily Rate (ADR) Distribution



Observation

The Average Daily Rate (ADR) shows noticeable variation across hotel bookings. Most reservations fall within a moderate price range, while a smaller number of bookings have significantly higher daily rates.

Interpretation

The distribution indicates that the majority of customers book rooms at standard prices. Higher ADR values are generally associated with premium rooms, peak travel seasons and luxury services.

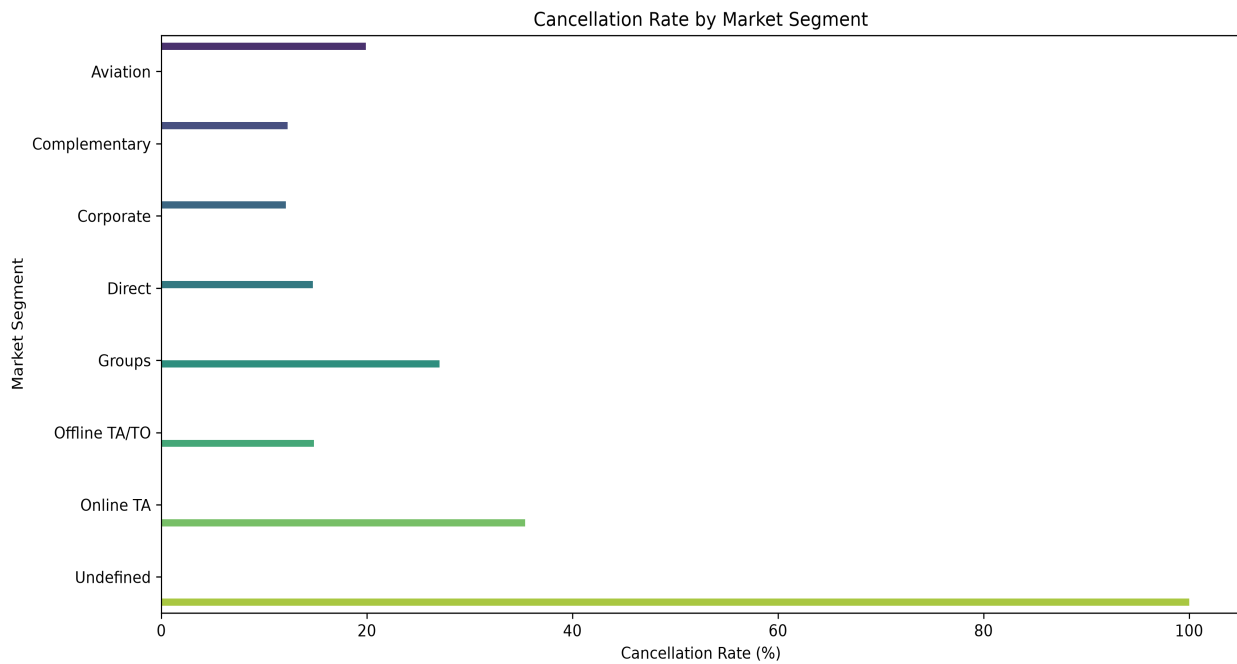
Business Insight

ADR is one of the most important hotel performance indicators. Monitoring ADR helps management evaluate pricing effectiveness, revenue generation and customer spending behaviour.

Recommendation

- Implement dynamic pricing based on seasonal demand.
- Promote premium room categories during high-demand periods.
- Offer attractive pricing packages during off-season periods.
- Continuously monitor ADR trends to maximize hotel revenue.

7. Cancellation Analysis



Observation

The cancellation analysis indicates that while a large proportion of bookings are successfully completed, a significant percentage of reservations are cancelled before the arrival date. This represents an important challenge for hotel operations and revenue management.

Interpretation

Booking cancellations reduce occupancy levels and create uncertainty in demand forecasting. Factors such as long lead times, flexible cancellation policies and changes in customer travel plans contribute to cancellation behaviour.

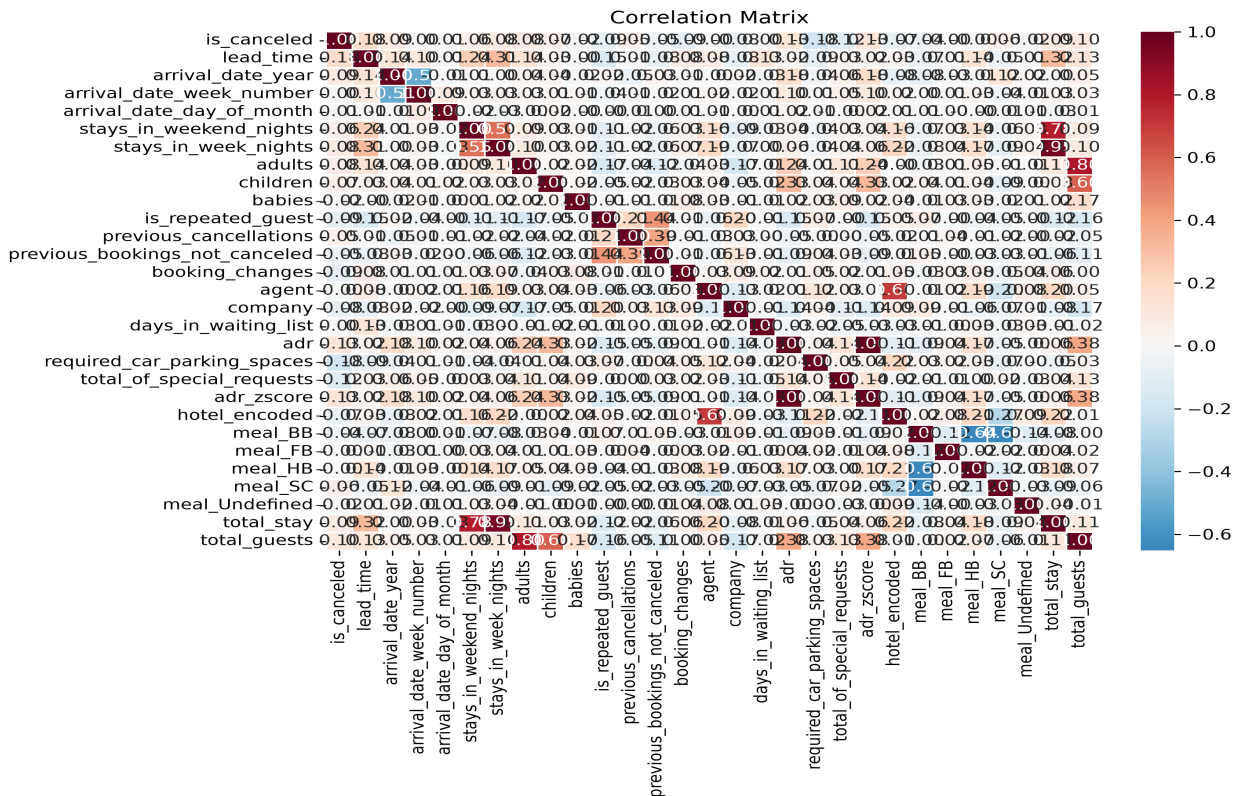
Business Insight

High cancellation rates negatively affect hotel revenue, resource planning and room utilization. Understanding customer cancellation patterns enables hotels to improve booking management strategies and increase occupancy.

Recommendation

- Identify customers with a higher probability of cancellation.
- Introduce attractive non-refundable or partially refundable booking options.
- Apply dynamic overbooking strategies based on historical cancellation trends.
- Send reminder emails and promotional offers before arrival to reduce last-minute cancellations.

8. Correlation Analysis



Observation

The correlation heatmap illustrates the strength and direction of relationships among numerical variables in the hotel booking dataset. Most variables exhibit weak to moderate correlations, while a few variables show stronger positive or negative relationships.

Interpretation

Correlation analysis helps identify variables that influence one another. Highly correlated features may provide valuable insights for predictive modeling, while weakly correlated variables may contribute independently to hotel booking behaviour.

Business Insight

Understanding relationships between variables helps hotel management identify the key drivers of booking demand, pricing, guest behaviour and operational performance. These insights support better strategic and business decisions.

Recommendation

- Use highly correlated variables for predictive analytics.
- Monitor relationships between pricing, lead time and cancellations to improve forecasting.
- Apply correlation analysis during feature selection for machine learning models.
- Continuously update correlation analysis as new booking data becomes available.

9. Overall Business Insights

The Exploratory Data Analysis provided several valuable insights into hotel booking patterns, customer behaviour and operational performance. These findings can assist hotel management in making strategic business decisions.

Key Business Insights

- City Hotels receive significantly more bookings than Resort Hotels, indicating higher demand for urban accommodations.
- Hotel demand varies across seasons, with peak travel periods generating the highest booking volumes.
- Most reservations are made by transient customers, highlighting the importance of individual travellers.
- Lead Time distribution shows that customers generally book closer to their arrival date, while a smaller number make reservations several months in advance.
- Average Daily Rate (ADR) varies across bookings, reflecting differences in room categories, seasonal pricing and customer preferences.
- Booking cancellations represent a considerable business challenge and directly affect hotel occupancy and revenue.
- Correlation analysis indicates that multiple variables contribute to hotel booking behaviour, supporting future predictive analytics and machine learning applications.
- Overall, customer behaviour, pricing strategy and seasonal demand are the primary drivers of hotel business performance.

10. Final Recommendations

Based on the Exploratory Data Analysis, the following recommendations are proposed to improve hotel operations, customer satisfaction and overall business performance.

Recommendations

- 1. Dynamic Pricing Strategy** Adjust room prices according to seasonal demand, occupancy levels and booking trends to maximize revenue.
- 2. Reduce Booking Cancellations** Introduce attractive non-refundable booking options, automated reminder emails and loyalty benefits to reduce last-minute cancellations.
- 3. Improve Occupancy During Off-Seasons** Launch promotional campaigns, discounted packages and family vacation offers to increase bookings during low-demand periods.
- 4. Strengthen Customer Retention** Develop loyalty programs, personalized offers and reward systems for repeat customers.
- 5. Optimize Resource Planning** Plan staffing levels, room inventory and hotel operations based on seasonal booking forecasts and historical demand.
- 6. Utilize Data-Driven Decision Making** Continuously monitor hotel performance using dashboards, KPIs and statistical analysis to support strategic business decisions.
- 7. Future Scope** The current analysis can be extended by developing machine learning models for booking cancellation prediction, customer segmentation, demand forecasting and dynamic pricing optimization.

Conclusion

The Hotel Booking Demand Analysis successfully explored customer booking behaviour, seasonal demand, pricing trends and operational performance. The insights obtained through Exploratory Data Analysis provide valuable support for hotel management in improving occupancy, revenue generation and customer satisfaction. This report demonstrates the practical application of Python, Statistics, Data Analytics and Business Intelligence techniques in solving real-world hospitality business problems.